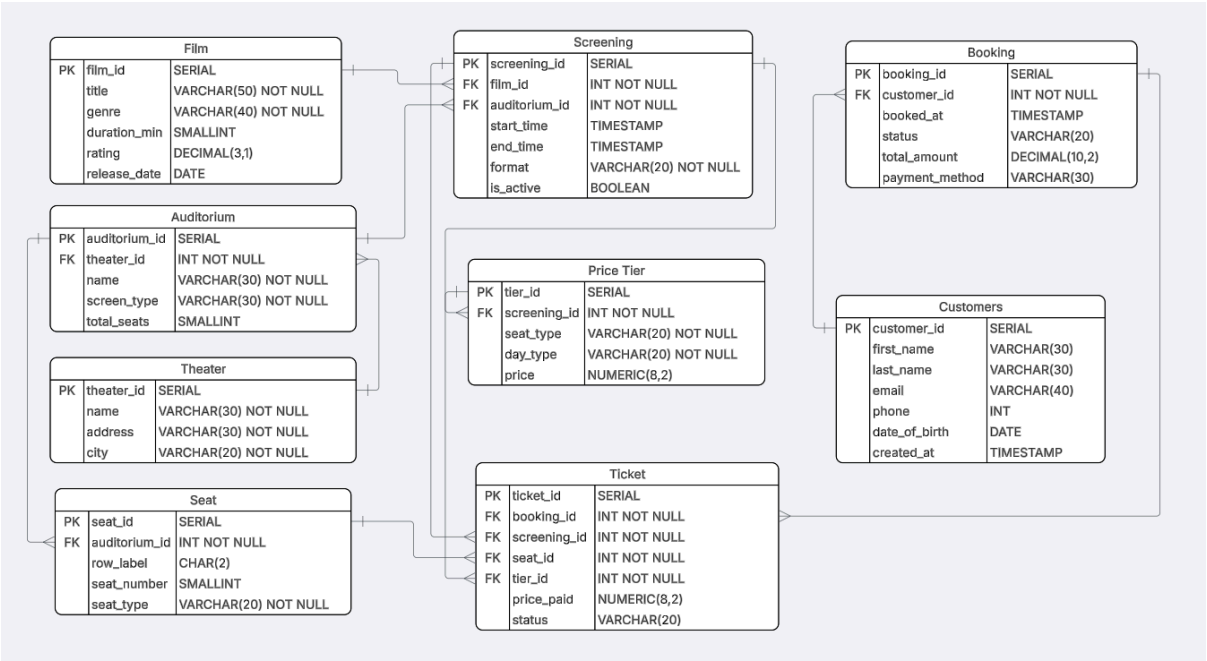
1.1 Logical Scheme



1.2 Objects

Table Description

The database is designed using related tables where main entities store general data, and other tables reference them through foreign keys (one-to-many relationships). Core tables such as Film, Theater, and Customers contain basic information, while dependent tables like Screening, Booking, and Ticket use this data to manage operations without duplication.

Table Name	Field name	Field Description	Data Type / Constraints
FIlm	film_id	the id of film	SERIAL, PK
	title	title of film	VARCHAR(50), NOT NULL
	genre	FIlm genre	VARCHAR(40) NOT NULL
	duration_min	Duration in minutes	SMALLINT
	rating	Rating of film	DECIMAL(3,1)
	release_date	Release date of film	DATE

Film and Screening (one-to-many).

Example with data

film_id	title	genre	duration_min	rating	release_date
1	Inception	Sci_Fi	148	8,8	2010-07-16
2	Avatar	Action	162	7,9	2009-12-18

Table Name	Field name	Field Description	Data Type / Constraints
Theater	theater_id	the id of theater	SERIAL, PK
	name	name of theater	VARCHAR(30), NOT NULL
	address	location of theater	VARCHAR(30) NOT NULL
	city	city where theater is located	VARCHAR(20) NOT NULL

Theater and Auditorium (one-to-many).

Example with data

theater_id	name	address	city
1	Cinema City	Main St 10	Budapest
2	Mega Cinema	Satpayev 25	Almaty

Table Name	Field name	Field Description	Data Type / Constraints
Auditorium	auditorium_id	the id of auditorium	SERIAL, PK
	theater_id	linked theater id	INT FK NOT NULL
	name	Auditorium name	VARCHAR(30) NOT NULL
	screen_type	Type (IMAX, 3D, etc.)	VARCHAR(30) NOT NULL
	total_seats	number of seats	SMALLINT

Auditorium and Theater (many-to-one), Auditorium and Seat (one-to-many), Auditorium and Screening (one-to-many).

Example with data

auditorium_id	theater_id	name	screen_type	total_seats
1	1	Hall 1	IMAX	100
2	1	Hall 2	3D	80

Table Name	Field name	Field Description	Data Type / Constraints
Seat	seat_id	The id of seat	SERIAL, PK
	auditorium_id	Linked auditorium	INT FK NOT NULL
	row_label	Row letter	CHAR(2) NOT NULL
	seat_number	Seat number	SMALLINT
	seat_type	Type (VIP, Regular)	VARCHAR(20) NOT NULL

Seat and Auditorium (many-to-one), Seat and Ticket (one-to-many).

Example with data

seat_id	auditorium_id	row_label	seat_number	seat_type
1	1	A	1	Regular
2	1	B	2	VIP

Table Name	Field name	Field Description	Data Type / Constraints
Screening	screening_id	The id of screening	SERIAL, PK
	film_id	Film being shown	INT FK NOT NULL
	auditorium_id	Auditorium	INT FK NOT NULL
	start_time	Start time of film	TIMESTAMP
	end_time	End time of film	TIMESTAMP
	format	Format(2D, 3D, IMAX)	VARCHAR(20) NOT NULL
	is_active	is screening active	BOOLEAN

Screening and Film (many-to-one), Screening and Auditorium (many-to-one), Screening and Ticket (one-to-many), Screening and Price_Tier (one-to-many).

Example with data

screening_id	film_id	auditorium_id	start_time	end_time	format	is_active
1	1	1	2026-04-15 18:00	2026-04-15 20:30	IMAX	TRUE
2	2	**2**	2026-04-15 21:00	2026-04-15 23:40	3D	FALSE

Table Name	Field name	Field Description	Data Type / Constraints
Price_Tier	tier_id	Unique tier ID	SERIAL, PK
	screening_id	Linked screening	INT FK NOT NULL
	seat_type	Seat category	VARCHAR(20), NOT NULL
	day_type	Weekday/Weekend	VARCHAR(20), NOT NULL
	price	Ticket price	NUMERIC(8,2)

Price_Tier and Screening (many-to-one), Price_Tier and Ticket (one-to-many).

Example with data

tier_id	screening_id	seat_type	day_type	price
1	1	Regular	Weekday	1200.00
2	1	**VIP**	Weekend	2000.00

Table Name	Field name	Field Description	Data Type / Constraints
Customers	customer_id	Unique customer ID	SERIAL, PK
	first_name	First name of customer	VARCHAR(30) NOT NULL
	last_name	Last name of customer	VARCHAR(30) NOT NULL
	email	Email of customer	VARCHAR(30) NOT NULL
	phone	Phone number	INT NOT NULL
	date_of_birth	Birth date	DATE
	created_at	Account creation time	TIMESTAMP

Customer and Booking (one-to-many).

Example with data

customer_id	first_name	last_name	email	phone	date_of_birth	created-at
1	Ilya	Osipov	monesy@mail.com	777111222	2005-05-01	2026-04-10 12:00
2	Danil	Kryshkovets	donk@mail.com	777333444	2007-01-25	2026-04-11 14:30

Table Name	Field name	Field Description	Data Type / Constraints
Booking	booking_id	Unique booking ID	SERIAL, PK
	customer_id	Linked customer	INT FK NOT NULL
	booked_at	Booking time	TIMESTAMP
	status	Booking status	VARCHAR(20), NOT NULL
	total_amount	Total price	DECIMAL(10,2)
	payment_method	Payment type	VARCHAR(30), NOT NULL

Booking and Customer (many-to-one), Booking and Ticket (one-to-many).

Example with data

booking_id	customer_id	booked_at	status	total_amount	payment_method
1	1	2026-04-12 15:00	confirmed	2400.00	card
2	1	2026-04-12 16:00	pending	1500.00	cash

Table Name	Field name	Field Description	Data Type / Constraints
Ticket	ticket_id	Unique ticket ID	SERIAL, PK
	booking_id	Linked booking	INT FK NOT NULL
	screening_id	Linked screening	INT, FK, NOT NULL
	seat_id	Selected seat	INT, FK, NOT NULL
	tier_id	Price tier	INT, FK, NOT NULL
	price_paid	Paid price	NUMERIC(8,2)
	status	Ticket status	VARCHAR(20)

Ticket and Booking (many-to-one), Ticket and Screening (many-to-one), Ticket and Seat (many-to-one), Ticket and Price_Tier (many-to-one).

Example with data

ticket_id	booking_id	screening_id	seat_id	tier_id	price_paid	status
1	1	1	1	1	1200.00	paid
2	1	1	2	1	1200.00	reserved